All questions in the exam have one (1) and only one correct alternative.

Answering with more than one alternative for a question will result in zero (0) points for the question.

General instruction for understanding the code

Unless the question states differently, understand the code in this exam as:

1. Javascript code
2. each question has its own javascript code file, that is run using node, e.g. think of the code for a question as: - its all in own file (`question8.js`), and - it is executed using the command `node question8.js`
3. the latest javascript standard.

Questions

Section 1: Javascript basics

1. What kind of language is JavaScript? `(1p)`

a) A compiled language
b) A hybrid language
c) An interpreted language

2. In JavaScript, which of the following are both types? `(1p)`

a) `Undefined` and `Null`
b) `null` and `NaN`

c) `undefined` and `NaN`

3. In JavaScript, the `this` keyword: (1p)

a) Is always lexically scoped
b) always refers to the global object
c) Is undefined in functions

4. In JavaScript, which declaration is function-scoped? (1p)

a) `let`
b) `const`
c) `var`

5. What allows functions in JavaScript to access variables from an outer function scope after the outer function has returned? (1p)

a) Generators
b) Closures
c) Scopes

6. In a browser, what object represents the webpage? (1p)

a) `window`
b) `page`
c) `document`

7. In JavaScript, what does JSON stand for? (1p)

a) Java Syntax Object Notation
b) JavaScript Object Notification
c) JavaScript Object Notation

8. What is the keyword for making a sub class in JavaScript? (1p)

a) `subclass`
b) `inherit`
c) `extends`

9. How is JavaScript's execution model described? (1p)

a) Multi-threaded
b) Dual-threaded
c) Single-threaded

10. Which of the following has the role of the global javscript object in a web browser environment? (1p)

a) `global`
b) `window`
c) `browser`

Section 2: Types and Type conversion

11. Given a variable that has the value:

```
undefined
```

What is the resulting value when converted to a *number*? (2p)

1. `undefined`
2. `null`
3. `true`
4. `false`
5. `""`
6. `"1.2"`
7. `1.2`
8. `"one"`
9. `0`
10. `1`
11. `Infinity`
12. `-Infinity`
13. `NaN`

12. Given a variable that has the value:

```
false
```

What is the resulting value when converted to a *number*? (2p)

1. undefined
2. null
3. true
4. false
5. “”
6. “1.2”
7. 1.2
8. “one”
9. 0
10. 1
11. Infinity
12. -Infinity
13. NaN

13. Given a variable that has the value:

```
undefined
```

What is the resulting value when converted to a *boolean*? (2p)

1. undefined
2. null
3. true
4. false
5. ""
6. "1.2"
7. 1.2
8. "one"
9. 0
10. 1
11. Infinity
12. -Infinity
13. NaN

14. Given a variable that has the value:

```
""
```

What is the resulting value when converted to a *boolean*? (2p)

1. undefined
2. null
3. true
4. false
5. ""
6. "1.2"
7. 1.2
8. "one"
9. 0
10. 1
11. Infinity
12. -Infinity
13. NaN

Section 3: Functions

15. Given the following JavaScript code:

```
function greet(name) {
    return "Hello, " + name + "!";
}
```

Which of the following will print "Hello, Alice!"? (2p)

a) `greet(Alice)`
b) `greet("Alice")`
c) `greet.name("Alice")`
d) `greet("Hello, Alice!")`

16. Which of the following is NOT a valid way to invoke a function in JavaScript? (2p)

a) `myFunction()`
b) `(myFunction)()`
c) `myFunction[]`
d) `myFunction.call()`

17. Given the following JavaScript function:

```
function displayInfo(name, age = 25) {
    return `${name} is ${age} years old.`;
}
```

What will the output be for `displayInfo("Bob")`? (2p)

a) `Bob is undefined years old.`
b) `Bob is 0 years old.`
c) `Bob is NaN years old.`
d) `Bob is 25 years old.`

18. Given the following JavaScript function:

```
function sum(...numbers) {
    return numbers.reduce((acc, curr) => acc + curr, 0);
}
```

Which two of the following will return the value `6`? (2p)

a) `sum([1, 2, 3])`
b) `sum(1, 2, 3)`
c) `sum(...[1, 2, 3])`
d) `sum({1, 2, 3})`

19. Consider the following code:

```
let x = function(y) {
    return y * y
}

let z = x
```

What will `z(4)` return? `(2p)`

a) `8`
b) `16`
c) `4`
d) `undefined`

Section 4: Exceptions, throw, try/catch/finally

20. What will the following code output? `(2p)`

```
try {
    console.log("A")
    throw new Error("An error occurred")
    console.log("B")
} catch (e) {
    console.log("C")
} finally {
    console.log("D")
}
```

a) A B C D
b) A B D
c) A C D
d) A D

21. Which error type will be thrown by the following code? `(2p)`

```
let obj = {}
console.log(obj.property.undefinedProperty)
```

a) ReferenceError
b) TypeError
c) SyntaxError
d) RangeError

Section 5: String

22. What will the following JavaScript code return? `(1p)`

```
"Web Development".endsWith("ment")
```

a) true
b) false
c) “Web Develop”
d) “ment”

23. What is the result of the following code? (1p)

```
let greeting = "   Hello World!   "
console.log(greeting.trim())
```

a) “Hello World!”
b) ” Hello World! ”
c) “HelloWorld!”
d) “Hello World! ”

24. How will the following code evaluate? (1p)

```
"JavaScript".charAt(5)
```

a) “S”
b) “r”
c) “a”
d) “c”

25. Consider the following JavaScript code:

```
let value = "25.50"
console.log(parseFloat(value) + 10)
```

What is the output? (1p) a) “35.50” b) 35.5 c) “2550” d) 255

Section 6: Loops (for, for/of, for/in, while, do/while)

26. What will be the output of the following code? (2p)

```
let sum = 0
for(let i = 1; i <= 4; i++){
    if(i % 2 == 0){
        continue
    }
    sum += i
}
console.log(sum)
```

a) 4
b) 5
c) 6
d) 7

27. What will be the output value of 'output' after the following code executes? (2p)

```
let arr = [1, 2, 3, 4, 5]
let output = ""
arr.forEach(function(num) {
    if(num % 2 === 0){
        return
    }
    output += num
})
console.log(output)
```

a) "12345"
b) "135"
c) "24"
d) ""

28. What will be the output of the following code? (2p)

```
outerLoop:
for(let i = 1; i <= 3; i++){
    for(let j = 1; j <= 3; j++){
        if(j === 2){
            break outerLoop
        }
        console.log(i, j)
    }
}
```

a) 1 1
b) 1 1 (newline) 1 2
c) 1 1 (newline) 2 1 (newline) 3 1
d) 1 1 (newline) 2 1 (newline) 3 1 (newline) 3 2

29. What is the output of the following code? (2p)

```
let i = 0
while(i++ < 5) {
    if(i === 3) {
        break
    }
}
console.log(i)
```

a) 2

b) 3
c) 5
d) 4

30. What's the output of the following code? (2p)

```
let numbers = [1, 2, 3, 4]
let output = []
numbers.forEach(num => {
    if(num % 2 == 0) {
        return
    }
    output.push(num)
})
console.log(output)
```

a) [0]
b) [1,3,4]
c) [1,3]
d) An error message

Section 7: Arrays and its functions

31. What will the following code output? (2p)

```
const arr = [2, 4, 6, 8]
console.log(arr[2])
```

a) 4
b) 6
c) 8
d) 2

32. What is the output of the following code? (2p)

```
const numbers = [1, 2, 3, 4, 5]
numbers.pop()
console.log(numbers)
```

a) [1, 2, 3, 4, 5]
b) [1, 2, 3, 4]
c) [2, 3, 4, 5]
d) []

33. What will the following code snippet output? (2p)

```
const values = [10, 20, 30, 40, 50]
console.log(values.slice(1, 4))
```

a) [10, 20, 30, 40]
b) [20, 30, 40, 50]
c) [20, 30, 40]
d) [10, 20, 30]

34. What will be the output of the following code? (2p)

```
const items = [1, 2, 3, 4, 5]
console.log(items.reverse())
```

a) [5, 4, 3, 2, 1]
b) [1, 2, 3, 4, 5]
c) 5
d) undefined

Section 8: Objects

35. What is the output of the code below? (2p)

```
const student = {
    grade: 90
}
console.log(student.grade)
```

a) “student”
b) “grade”
c) 90
d) undefined

36. What is the output of the code below? (2p)

```
const city = {
    name: "London"
}
console.log("London" in city)
```

a) true
b) false
c) “London”
d) “population”

37. What is the output of the code below? (2p)

```
const colors = {
    red: "#FF0000",
    blue: "#0000FF"
}
let output = ""
for (let color in colors) {
    output += color
}
console.log(output)
```

a) “#FF0000#0000FF”
b) “redblue”
c) “red#FF0000blue#0000FF”
d) “colors”

38. What is the output of the code below? (2p)

```
const shape = {
    sides: 4,
    type: "square"
}
console.log(Object.keys(shape)[1])
```

a) 4
b) “square”
c) “sides”
d) “type”

39. What is the output of the code below? (2p)

```
const instrument = {
    type: "Guitar",
    strings: 6
}
console.log(Object.values(instrument)[1])
```

a) “Guitar”
b) 6
c) “type”
d) “instrument”

40. What is the output of the code below? (2p)

```
const team = {
    name: "Warriors",
    shout: function() {
    return "Go " + this.name + "!"
    }
}
const cheer = team.shout
console.log(cheer.call(team))
```

a) "Go Warriors!"
b) "Go !"
c) "Go this.name!"
d) "Go team!"

Section 9: Prototypes and classes

41. What is the output of the code below? (2p)

```
function Bird(name) {
    this.name = name
}

Bird.prototype.fly = function() {
    return `${this.name} flies`
}

function Penguin(name) {
    Bird.call(this, name)
}

Penguin.prototype = Object.create(Bird.prototype)
Penguin.prototype.constructor = Penguin

Penguin.prototype.fly = function() {
    return `${this.name} can't fly`
}

let happyFeet = new Penguin("Happy Feet")

console.log(happyFeet.fly())
```

a) "Happy Feet flies"
b) "Happy Feet can't fly"
c) "Bird flies"

d) “Penguin flies”

42. What is the output of the code below? (2p)

```
function Car(model) {
    this.model = model
}

Car.prototype.drive = function() {
    return `${this.model} is driving`
}

let car1 = new Car("Sedan")
let car2 = Object.create(Car.prototype)
car2.model = "SUV"

console.log(car2.drive())
```

a) “Sedan is driving”
b) “SUV is driving”
c) “Car is driving”
d) Error

43. What is the output of the code below? (2p)

```
class Square {
    constructor(side) {
        this.side = side
    }

    area() {
        return this.side * this.side
    }
}

let squareObj = new Square(4)
console.log(squareObj.area())
```

a) 16
b) “Square with side 4”
c) 8
d) Error

Section 10: Iterators & generators

44. What is the output of the code below? (2p)

```
let iterable = [3, 4, 5]
let iterator = iterable.entries()

let firstOutput = iterator.next().value
console.log(firstOutput)
```

a) [0, 3]
b) [1, 4]
c) 3
d) [3, 0]

45. What will be the output of the following code? (2p)

```
let colors = ['red', 'blue', 'green']
let iterator = colors.keys()

let secondKey = iterator.next().value
secondKey = iterator.next().value
console.log(secondKey)
```

a) 'red'
b) 1
c) 'blue'
d) 0

46. What will be the output of the following code? (2p)

```
function* squareGenerator() {
    yield 2 * 2
    yield 3 * 3
    yield 4 * 4
}

let gen = squareGenerator()
let firstSquare = gen.next().value
console.log(firstSquare)
```

a) 4
b) 9
c) 16
d) 2

47. What will be the output of the following code? (2p)

```
function* fibonacci() {
    let [prev, curr] = [0, 1]
    while (true) {
        [prev, curr] = [curr, prev + curr]
        yield curr
    }
}

let sequence = fibonacci()
let thirdNum = sequence.next().value
thirdNum = sequence.next().value
thirdNum = sequence.next().value
console.log(thirdNum)
```

a) 1
b) 2
c) 3
d) 5

Section 11: Modules

48. Consider the following files:

greetModule.js:

```
const greeting = "Hello, World!"
export default greeting
```

main.js:

```
import greet from './greetModule.js'

console.log(greet)
```

What will be the output when `main.js` is executed? (2p)

a) `Hello, World!`
b) `greeting`
c) `greetModule`
d) `undefined`

Section 12: Std library

49. What will be the output of the following code? (1p)

```
const numbers = [1, 2, 3]
const doubled = numbers.map(x => x * 2)
console.log(doubled)
```

a) [1, 2, 3]
b) 2, 4, 6
c) [2, 4, 6]
d) undefined

50. What will be the output of the following code? (1p)

```
const values = [1, 2, 3, 4]
const sum = values.reduce((acc, curr) => acc + curr, 0)
console.log(sum)
```

a) 10
b) [1, 2, 3, 4]
c) 0
d) [10]

51. What will be the output of the following code? (1p)

```
const array = [10, 15, 20, 25, 30]
const result = array.filter(item => item > 20)
console.log(result)
```

a) [10, 15, 20, 25, 30]
b) [25, 30]
c) [10, 15, 20]
d) [20, 25, 30]

52. What will be the output of the following code? (1p)

```
const letters = ['a', 'b', 'c']
let output = ''
letters.forEach(letter => output += letter.toUpperCase())
console.log(output)
```

a) 'ABC'
b) 'abc'
c) ['A', 'B', 'C']
d) 'aBc'

Section 13: HTML & Script

53. Given the following HTML with JavaScript code:

```
<div id="message">Hello, World!</div>

<script>
    const divElement = document.getElementById("message")
    divElement.innerText = "Greetings, Earth!"
</script>
```

What will be displayed on the web page? `(1p)`

a) Hello, World!
b) Greetings, Earth!
c) [object HTMLElement]
d) An error will be thrown.

54. Given the following HTML with JavaScript code:

```
<button id="btn">Click me!</button>

<script>
    const button = document.getElementById("btn")
    button.addEventListener('click', function() {
    alert("Button was clicked!")
    });
</script>
```

What happens when the button is clicked? `(1p)`

a) The page will reload.
b) An alert will pop up with the message “Button was clicked!”.
c) The text of the button changes to “Button was clicked!”.
d) Nothing happens.

55. Given the following HTML with JavaScript code:

```
<input type="text" id="nameInput" value="John Doe">

<script>
    const inputElement = document.getElementById("nameInput")
    console.log(inputElement.value)
</script>
```

What will be printed to the console? `(1p)` a) John Doe b) [object HTMLInputElement] c) undefined d) nameInput

56. Given the following HTML with JavaScript code:

```
<p id="highlighted" style="color: blue;">This is a text.</p>

<script>
    const paragraph = document.getElementById("highlighted");
    paragraph.style.color = "red";
</script>
```

What color will the text inside the paragraph be? (1p)

a) Blue
b) Red
c) Green
d) Black

Section 14: Some more code...

Consider this code for the next five (5) questions.

```
class Book {
  constructor(title, author, pages) {
    this.title = title;
    this.author = author;
    this.pages = pages;
    this.readCount = 0;
  }

  read() {
    this.readCount++;
  }

  description() {
    return `${this.title} by ${this.author} has ${this.pages} pages. Read count: ${this.read
  }
}

const myBook = new Book('The Art of Coding', 'John Doe', 250);
myBook.read();

console.log(myBook.description());
```

57. What will be the output when the above code is executed? (2p)

a) 'The Art of Coding by John Doe has 250 pages. Read count: 0'
b) 'The Art of Coding by John Doe has 250 pages. Read count: 1'
c) 'The Art of Coding by John Doe has 250 pages.'

d) 'John Doe has written The Art of Coding with 250 pages.'

58. Which method is used to increment the `readCount` property of a book? (2p)

a) `increaseReadCount()`
b) `description()`
c) `read()`
d) `addReadCount()`

59. After the following lines of code:

```
myBook.read();
myBook.read();
console.log(myBook.description());
```

What will be the output? (2p)

a) 'The Art of Coding by John Doe has 250 pages. Read count: 1'
b) 'The Art of Coding by John Doe has 250 pages. Read count: 2'
c) 'The Art of Coding by John Doe has 250 pages. Read count: 3'
d) 'The Art of Coding by John Doe has 250 pages. Read count: 0'

60. What does the `constructor` method in the `Book` class do? (2p)

a) Initializes properties of the object when it is created.
b) Provides a description of the book.
c) Increments the read count of the book.
d) Prints the book details.

61. If we want to get the number of times the `myBook` object has been read without printing the entire description, which property should we access? (2p)

a) `myBook.title`
b) `myBook.pages`
c) `myBook.readCount`
d) `myBook.author`